

Foundations of Audio Signal Processing

Exercise sheet 5

To be uploaded in eCampus till: 21-11-2025 22:00 (strict deadline)

Exercise 5.1

[6 points]

Let $f : \mathbb{R} \mapsto \mathbb{C}$ be defined as follows:

$$f(t) = e^{-\alpha|t|}$$

Find the values of $\alpha \in \mathbb{R}$ for which it holds that $f(t) \in L^2(\mathbb{R})$.

Exercise 5.2

[6 points]

For a positive parameter $c \in \mathbb{R}$ define the function $f_c : \mathbb{R} \rightarrow \mathbb{R}$

$$f_c(t) := \begin{cases} \frac{1}{c}t + 1 & \text{for } t \in [-c, 0], \\ -\frac{1}{c}t + 1 & \text{for } t \in (0, c], \\ 0 & \text{otherwise.} \end{cases}$$

Determine c , such that f_c is normed with respect to the L^1 -Norm (i.e. $\|f_c\|_1 = 1$).

Exercise 5.3

[3 + 3 + 3 = 9 points]

Classify the following DT signals by specifying all $p \in [1, \infty]$ with $x \in \ell^p(\mathbb{Z})$.

(a) $x(n) := e^n$.

(b) $x(n) := e^{in}$.

(c) $x(n) := \frac{1}{\sqrt{n}}, \quad n > 0$